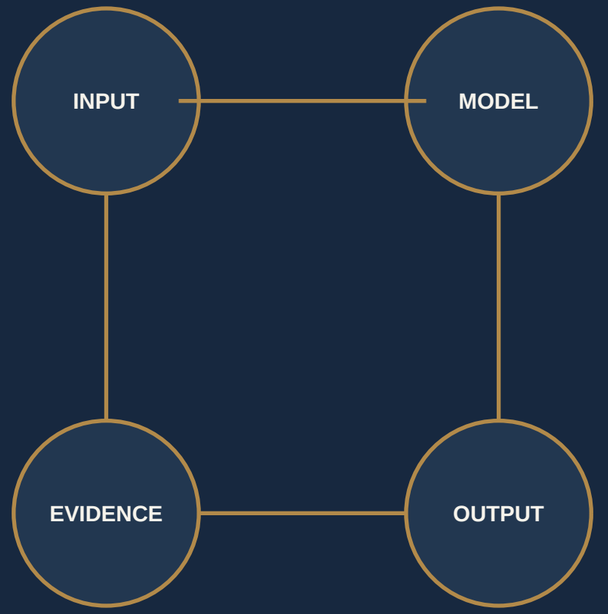
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Embedded Systems

AI.109 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define microcontrollers.
- Apply key methods including processors.
- Build or analyze a practical system involving memory.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Microcontrollers

02

METHODS

Processors

03

SYSTEMS

Memory

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Microcontrollers

Microcontrollers

02 · Processors

Processors

03 · Memory

Memory

04 · GPIO

GPIO

05 · Timers

Timers

06 · Interrupts

Interrupts

07 · Communication protocols

Communication protocols

08 · Sensors

Sensors

Core ideas, made explicit

AI.109 • Distinguish definitions, assumptions and evidence

- Microcontrollers
- Processors
- Memory
- GPIO
- Timers

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

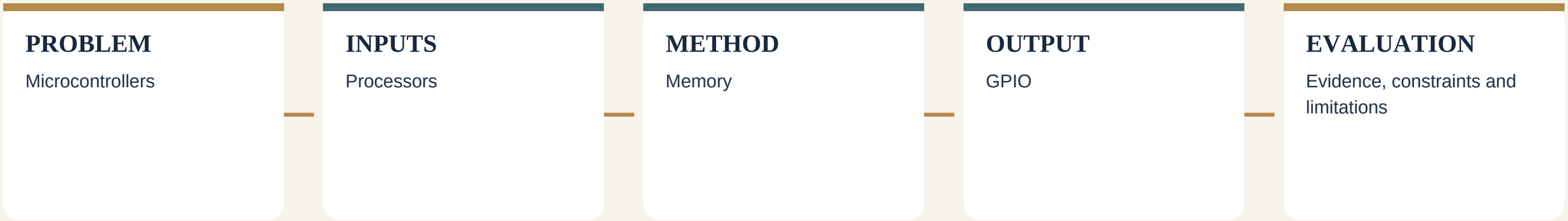
AI.109 • Distinguish definitions, assumptions and evidence

- Interrupts
- Communication protocols
- Sensors
- Actuators
- Real-time systems and embedded programming

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.



Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving microcontrollers.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using microcontrollers.
- Implement or simulate a baseline using processors.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.